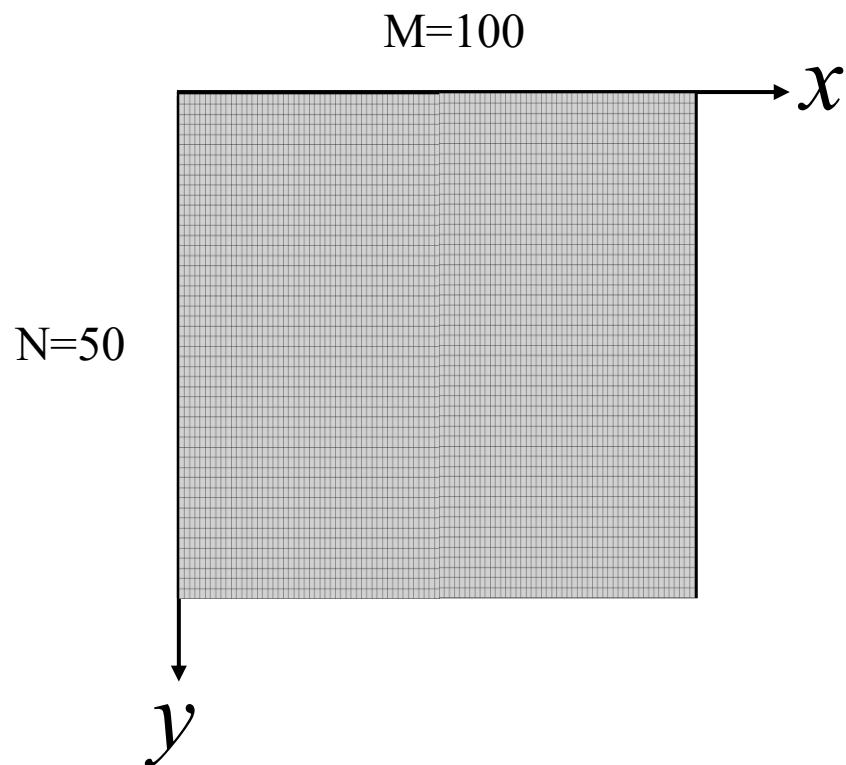


範例一:影像索引2D



Q: ➤ 已知索引值 $\alpha = 120$, 求 (x, y) 座標
➤ Given the index value α , find (x, y) .

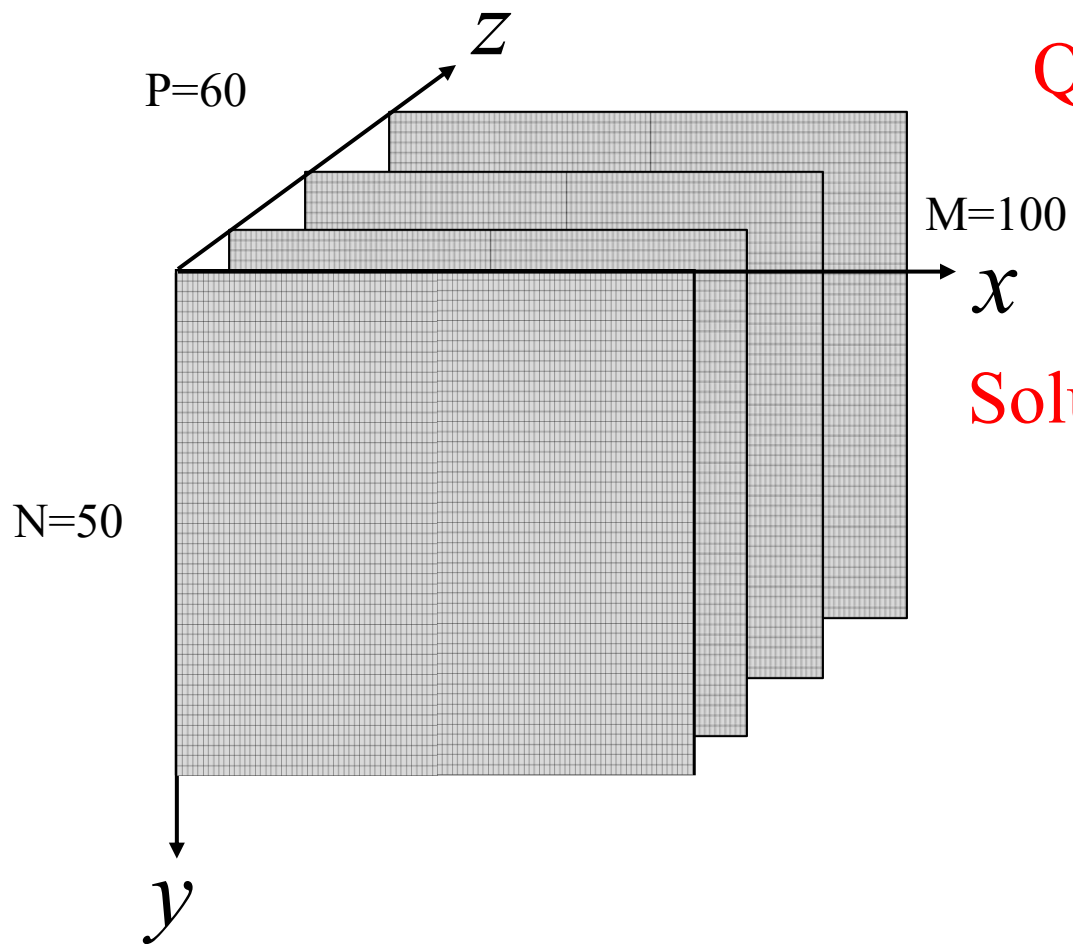
Solution:

$$\begin{aligned} x &= \alpha \bmod M \\ y &= \alpha // M \text{ (取商數)} \end{aligned}$$

$$\begin{aligned} x &= 120 \bmod 100 = 20 \\ y &= 120 // 100 = 1 \end{aligned}$$

Ans: $(x, y) = (20, 1)$ #

範例二:影像索引3D



- Q:**
- 已知索引值 $S = 5090$, 求 (x, y, z) 座標
 - Given the index value $S = 5090$, find (x, y, z) .

Solution:

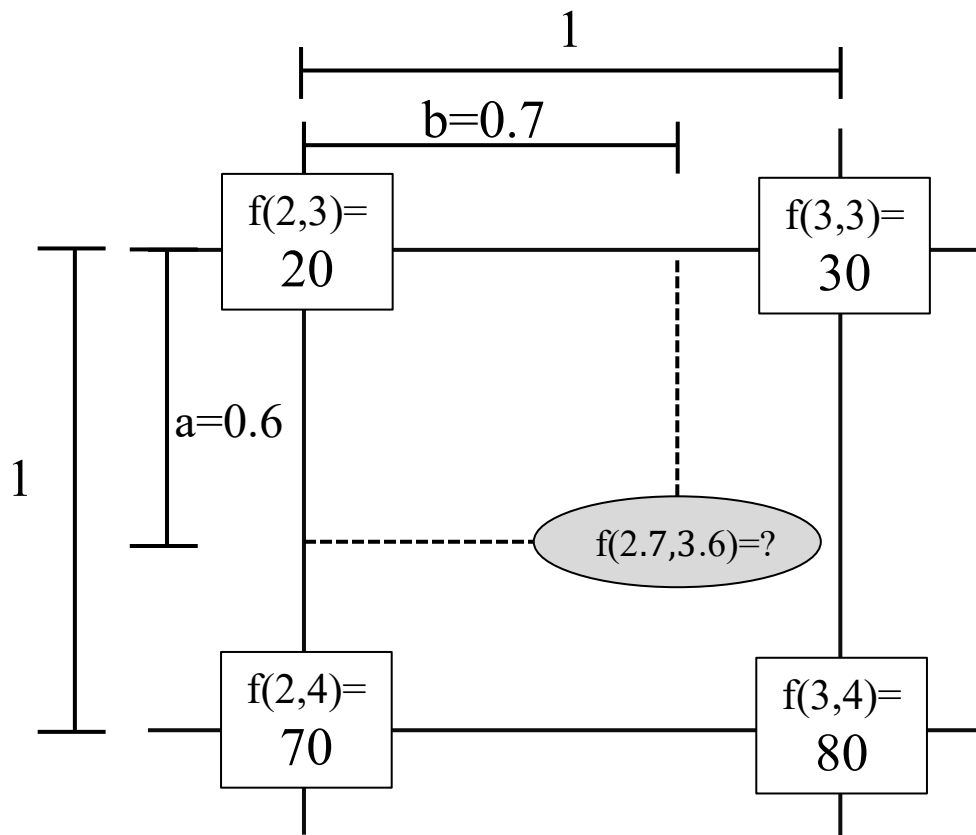
$$\begin{aligned}x &= S \bmod MN \bmod M \\y &= S \bmod MN // M \text{ (取商數)} \\z &= S // MN \text{ (取商數)}\end{aligned}$$

$$\begin{aligned}x &= 5090 \bmod (100 * 50) \bmod 100 = 90 \\y &= 5090 \bmod (100 * 50) // 100 = 0 \\z &= 5090 // (100 * 50) = 1\end{aligned}$$

Ans: $(x, y, z) = (90, 0, 1)$ #

範例三:雙線性內插(Bilinear interpolation)

- Q:** ➤ 利用雙線性內插法，計算 $f(2.7, 3.6)$ 的影像強度。
- Use bilinear interpolation to compute $f(2.7, 3.6)$ given the four corner values.



Sol:

$$b_1 = (1 - b) \times a_1 + b \times a_2$$

$$b_5 = (1 - b) \times a_3 + b \times a_4$$

$$b_3 = (1 - a) \times b_1 + a \times b_5$$

$$0.3 \times 20 + 0.7 \times 30 = 27$$

$$0.3 \times 70 + 0.7 \times 80 = 77$$

$$0.4 \times 27 + 0.6 \times 77 = 57$$

Ans: $f(2.7, 3.6) = 57$ #